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MINISTRY OF HIGHER EDUCATION

INTERNSHIP REPORT

WEB PLATFORM FOR RECRUITMENT AND JOB APPLICATION TRACKING

TALENTFLOW – Job Posting, Secure CV Management & Hiring Pipeline

Internship carried out at AFRILAND FIRST BANK, Douala, Bonanjo

From July 1 to August 31 2026, in preparation for advancement to 3rd year

Field: BACHELOR OF ENGINEERING (DATA SCIENCE AND ARTIFICIAL INTELLIGENCE).

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DEDICATION

This work is dedicated to my beloved family,
for their love, their sacrifices and their unfailing support.

ACKNOWLEDGEMENTS

Every accomplishment is the result of collaboration, guidance and perseverance. The completion of this internship report marks an important milestone in my academic and professional journey, and I am deeply grateful to all those whose support, advice and encouragement made it possible.

First and foremost, I give thanks to Almighty God for granting me life, health, strength and wisdom throughout this internship period.

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LIST OF ACRONYMS

Acronym	Meaning
API	Application Programming Interface
ATS	Applicant Tracking System
AES-GCM	Advanced Encryption Standard – Galois/Counter Mode
BPM	Business Process Management
BPMN	Business Process Model and Notation
CDC	Cahier Des Charges (requirements specification)
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
CV	Curriculum Vitae
DSI	Direction des Systèmes d'Information
HR	Human Resources
HTTPS	HyperText Transfer Protocol Secure
IUC	Institut Universitaire de la Côte
JSON	JavaScript Object Notation
JWT	JSON Web Token
MIME	Multipurpose Internet Mail Extensions
ORM	Object-Relational Mapping
OWASP	Open Worldwide Application Security Project
RBAC	Role-Based Access Control
REST	REpresentational State Transfer
SEAS	School of Engineering and Applied Sciences
SME	Small and Medium-sized Enterprise
SMTP	Simple Mail Transfer Protocol
SPA	Single Page Application
SQL	Structured Query Language
TLS	Transport Layer Security
UML	Unified Modeling Language
UUID	Universally Unique Identifier

ABSTRACT

This report presents the work carried out during my academic internship at **Afriland First Bank**, Douala – Bonanjo, from July 1 to August 31, 2026, in partial fulfilment of the requirements of the Bachelor of Engineering programme of the School of Engineering and Applied Sciences (SEAS) of the Institut Universitaire de la Côte (IUC).

During the internship, I observed that recruitment, both in the host structure and in the small and medium-sized enterprises (SMEs) that form a large part of its customers, is still handled with tools that were not designed for it: CVs arrive by e-mail, WhatsApp, paper and job boards, they are copied into Excel sheets and exchanged by e-mail, and nobody has a complete view of where each application stands. CVs are lost or duplicated, candidates rarely receive an answer, and personal data are stored without protection on personal phones and shared computers.

To address this problem, I designed and developed **TalentFlow**, a web platform for recruitment and job application tracking. The platform allows a company to publish job offers on a public career page, to receive CVs through a single secured channel, to model its own recruitment process as customizable stages displayed on a Kanban board, to search and filter candidates, and to follow its activity on an HR dashboard with automatic notifications. The solution is built with **React** for the front-end, **Node.js/Express** for the REST API and **PostgreSQL** for data storage. Particular attention was paid to the modelling of the recruitment business process and to the secure management of CVs: file validation and antivirus scanning, AES-256 encryption at rest, short-lived signed links, role-based access control and an audit trail of every access.

Keywords: recruitment, applicant tracking system, business process modelling, secure file management, React, Node.js, Express, PostgreSQL.

RÉSUMÉ

Ce rapport présente le travail réalisé au cours de mon stage académique effectué à **Afriland First Bank**, Douala – Bonanjo, du 1er juillet au 31 août 2026, dans le cadre de la formation de Bachelor of Engineering de la School of Engineering and Applied Sciences (SEAS) de l'Institut Universitaire de la Côte (IUC).

Au cours de ce stage, nous avons constaté que le recrutement, aussi bien dans la structure d'accueil que dans les petites et moyennes entreprises (PME) qui constituent une part importante de sa clientèle, est encore géré avec des outils inadaptés : les CV arrivent par e-mail, WhatsApp, papier et sites d'emploi, ils sont recopiés dans des fichiers Excel, et personne n'a une vue complète de l'état de chaque candidature. Des CV sont perdus ou dupliqués, les candidats reçoivent rarement une réponse et les données personnelles sont conservées sans protection.

Pour répondre à ce problème, nous avons conçu et développé **TalentFlow**, une plateforme web de recrutement et de suivi des candidatures. Elle permet de publier des offres d'emploi sur une page carrière publique, de recevoir les CV par un canal unique et sécurisé, de modéliser son propre processus de recrutement sous forme d'étapes personnalisables affichées dans un tableau Kanban, de rechercher et filtrer les candidats, et de piloter l'activité grâce à un tableau de bord RH et à des notifications automatiques. La solution repose sur **React**, **Node.js/Express** et **PostgreSQL**. Une attention particulière a été portée à la modélisation du processus métier et à la gestion sécurisée des CV : validation et analyse antivirus des fichiers, chiffrement AES-256, liens signés à durée limitée, contrôle d'accès par rôles et journalisation de chaque consultation.

Mots-clés : recrutement, suivi des candidatures, modélisation de processus métier, gestion sécurisée des fichiers, React, Node.js, Express, PostgreSQL.

GENERAL INTRODUCTION

Context of the study

Recruitment is one of the most decisive processes of any organisation: the quality of the people hired determines the quality of the services delivered. In Cameroon, as in many developing economies, small and medium-sized enterprises (SMEs) represent the large majority of companies and of formal jobs. These companies hire regularly but rarely have a dedicated Human Resources (HR) department or a recruitment software. Job offers are shared on Facebook, LinkedIn, WhatsApp groups or posted at the entrance of the premises, and applications arrive through all these channels at the same time.

This study was carried out at Afriland First Bank, Douala – Bonanjo, from July 1 to August 31, 2026. As a bank, Afriland First Bank works daily with SMEs, which are among its main customers, and it also recruits staff and interns itself. Through discussions with the HR staff and with customer advisers, I observed that the same difficulty appears everywhere: **applications are received through several channels and their follow-up is lost**.

Definition of the main concepts

- **Recruitment:** the set of activities by which an organisation attracts, selects and hires the people it needs (definition of the need, publication of the offer, collection of applications, selection, decision, integration).
- **Application (candidature):** the act by which a person applies for a given job offer; it is made of personal data, a CV and possibly answers to screening questions.
- **Recruitment pipeline:** the ordered sequence of stages (e.g. New, Screening, Interview, Test, Offer, Hired) through which each application moves until a final decision.
- **Applicant Tracking System (ATS):** a software application that centralises job offers and applications and supports the recruitment pipeline.
- **Business process modelling:** the representation of the activities, actors, rules and events of a business process in order to analyse and automate it.
- **Secure CV management:** the set of technical and organisational measures that guarantee the confidentiality, integrity, availability and lawful processing of candidates' personal data.

Problem statement

The central problem of this study can be stated as follows: **how can a small company centralise applications received from multiple channels, follow each of them through a recruitment process adapted to its own practices, and protect the personal data contained in the CVs?** This main question leads to the following specific questions:

- How can the different sources of applications be gathered into a single channel?
- How can a recruitment process be modelled so that each company can customise its own stages?

- How can HR staff quickly find the right candidates among hundreds of CVs?
- How can candidates and recruiters be kept informed without additional manual work?
- How can CVs be stored and consulted securely, in accordance with personal data protection rules?

Aim of the study

The general objective of this work is to design and implement a web platform for recruitment and job application tracking, named **TalentFlow**. More specifically, the platform must make it possible to: publish job offers and receive CVs; define customizable recruitment stages; filter and search candidates; and provide notifications and an HR dashboard. For the host company and its SME customers, the expected benefits are a reduction of the time spent on administrative tasks, no more lost applications, a better experience for candidates and a lawful, secured handling of personal data.

Organisation of the work

Apart from this general introduction and the general conclusion, this report is organised into two parts. **Part One (conceptual framework)** contains Chapter 1, which presents the host company and the activities carried out during the internship, and Chapter 2, which gives an overview of recruitment management and of the existing solutions. **Part Two (empirical study)** contains Chapter 3, which diagnoses the existing system and presents the requirements specification (cahier des charges) of the new platform, and Chapter 4, which presents the choice and the implementation of the solution, its installation procedure and its cost estimate.

PART ONE: CONCEPTUAL FRAMEWORK

Chapter 1: Presentation of the company and activities carried out during the internship period

Chapter 2: Overview of recruitment management and applicant tracking systems

CHAPTER 1: PRESENTATION OF THE COMPANY AND ACTIVITIES CARRIED OUT DURING THE INTERNSHIP PERIOD

Introduction

This chapter presents the host structure, Afriland First Bank, then describes week by week the activities carried out during the internship, and finally explains the choice and the justification of the research topic.

1.1 Presentation of the company

1.1.1 Identification sheet

Table 1: Identification sheet of Afriland First Bank

Item	Information
Company name	Afriland First Bank
Former name	CCEI Bank (Caisse Commune d'Épargne et d'Investissement)
Legal form	Public limited company (Société Anonyme)
Year of creation	1987
Sector of activity	Banking and financial services
Head office	Yaoundé, Cameroon
Internship site	Douala – Bonanjo
Group	Afriland First Group
Website	www.afrilandfirstbank.com

1.1.2 Historical background

Afriland First Bank was created in 1987 under the name **CCEI Bank** (Caisse Commune d'Épargne et d'Investissement) by Cameroonian promoters, with the objective of offering banking services adapted to local entrepreneurs. The bank took the name **Afriland First Bank** in 2002. Over the years it has developed a network of branches across the Cameroonian territory and has extended its activities to other African countries within Afriland First Group. It is today one of the leading commercial banks in Cameroon.

1.1.3 Missions and activities

As a commercial bank, Afriland First Bank offers a wide range of services to individuals, SMEs, large companies and institutions:

- collection of deposits and management of current and savings accounts;
- granting of loans to individuals and to companies, with a strong focus on SMEs and entrepreneurs;
- national and international money transfers and trade finance;

- ➢ electronic banking: cards, mobile and internet banking;
- ➢ advice and support to companies in the management of their activities.

1.1.4 Organisational chart

The bank is organised around a general management and functional directorates. Figure 1 gives a simplified view of the organisation, showing the unit in which the internship took place.

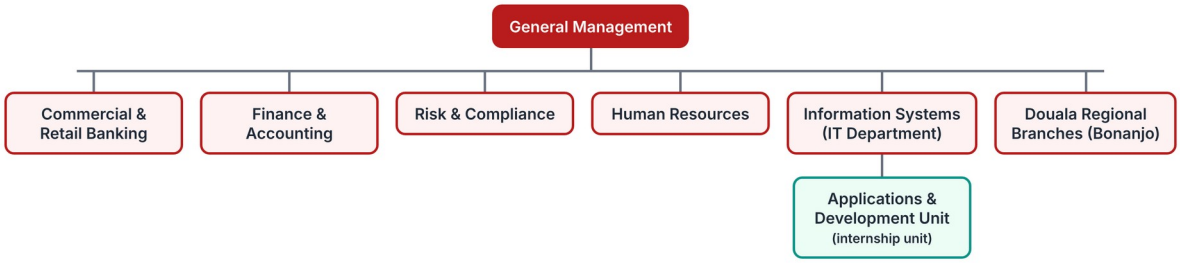


Figure 1: Simplified organisational chart of the host structure

1.1.5 The Information Systems department

The Information Systems department is responsible for the availability and the security of the information system of the bank: core banking applications, networks, workstations, electronic banking channels and internal business applications. Within this department, the applications and development unit designs or adapts internal tools for the other departments (Human Resources, Commercial, Risk, etc.). It is in this unit that I was received, which allowed me to work both with developers and with the business users of the HR department.

1.2 Activities carried out during the internship

1.2.1 Reception within the structure

On July 1, 2026, I was received by my professional supervisor, who presented the bank, its organisation and its internal rules (working hours, dress code, confidentiality, use of IT resources). I signed a confidentiality commitment and received access to a workstation and to the development environment of the unit.

1.2.2 Weekly activities

Table 2 summarises the activities carried out each week and what I learned from them.

Table 2: Weekly activities carried out during the internship

Week	Activities	What I learned / problems solved
W1 01 – 05 July	Reception, presentation of the bank and of the IT department; installation of the work environment; observation of the daily tasks of the unit.	Organisation of a bank, security rules, use of the ticketing tool.

Week	Activities	What I learned / problems solved
W2 06 – 12 July	Support to the IT help desk (workstation configuration, network issues); interviews with HR staff about the recruitment of interns and staff.	Troubleshooting methods; understanding of the recruitment process and of its pain points.
W3 13 – 19 July	Study of the existing process: collection of examples of Excel tracking sheets, e-mails and paper CVs; interviews with SME customers via account managers; writing of the requirements specification.	Requirements elicitation; writing of functional and non-functional requirements.
W4 20 – 26 July	UML modelling (use cases, classes, activities, sequences); design of the database; validation of the design with the supervisor.	Business process modelling; relational database design and normalisation.
W5 27 July – 02 Aug.	Set-up of the project (Git, Docker, PostgreSQL); development of authentication, users, roles and job offers in the Express API.	JWT authentication, password hashing, REST API design, Prisma migrations.
W6 03 – 09 Aug.	Development of the customizable pipeline and of the application workflow; development of the public career page in React.	Transactions in PostgreSQL, React components and state management.
W7 10 – 16 Aug.	Secure CV upload (validation, antivirus, encryption, signed URLs); candidate search with full-text search; notifications by e-mail and in-app.	File upload security (OWASP), PostgreSQL full-text search, job queues.
W8 17 – 23 Aug.	HR dashboard; unit and integration tests; deployment on a test server; demonstration to HR staff and collection of feedback.	Testing with Jest and Supertest; Docker deployment; user feedback.
W9 24 – 31 Aug.	Corrections after feedback; user guide and technical documentation; writing of the internship report; end-of-internship presentation.	Technical writing and oral presentation.

1.2.3 Tools at my disposal

Table 3: Tools used during the internship

Category	Tools
Hardware	Laptop (Intel Core i5, 16 GB RAM, 512 GB SSD), local network and Internet access
Operating systems	Windows 11 (workstation), Ubuntu Server 22.04 (test server)
Development	Visual Studio Code, Node.js 20, npm, Git and GitHub, Docker Desktop, Postman
Database	PostgreSQL 16, pgAdmin 4, Prisma ORM
Design	draw.io / StarUML (UML diagrams), Figma (interface mock-ups)
Office	Microsoft Word, Excel, PowerPoint, Microsoft Teams

1.3 Choice and justification of the research topic

During the second week, the HR staff explained that, for each internship or job campaign, they receive several hundred applications by e-mail, by WhatsApp and on paper. These applications are recopied in Excel files, CVs are forwarded by e-mail to managers, and it often becomes impossible to know which candidate has been contacted, interviewed or rejected. Account managers confirmed that many SME customers of the bank face exactly the same situation, with even fewer resources.

The topic **“Web platform for recruitment and job application tracking”** was therefore chosen because it answers a real and recurrent need of the host structure and of its customers, because its result can be used immediately, and because it allowed me to apply the skills of my training: requirements analysis, business process modelling, web development and information security.

Conclusion

This chapter presented Afriland First Bank, the activities carried out during the nine weeks of the internship and the reasons for the choice of the topic. The next chapter presents the theoretical background of recruitment management and the existing solutions.

CHAPTER 2: OVERVIEW OF RECRUITMENT MANAGEMENT AND APPLICANT TRACKING SYSTEMS

Introduction

This chapter presents the theory needed to understand the study: the recruitment process, the notion of business process modelling, the principles of personal data protection applied to CVs and the technologies of the web. It then compares the possible solutions to the problem.

2.1 The recruitment process

Whatever the size of the company, a recruitment generally follows the same main phases:

1. **Definition of the need:** job title, missions, required profile, contract type, budget;
2. **Sourcing:** publication of the offer on one or several channels (career page, job boards, social networks, schools, word of mouth);
3. **Collection of applications:** reception of CVs, cover letters and answers to questions;
4. **Selection:** screening of CVs, phone interview, tests, face-to-face interviews, reference checks;
5. **Decision and offer:** choice of the candidate, negotiation, offer letter;
6. **Integration (onboarding):** welcome and induction of the new employee;
7. **Closure:** answer to all the candidates who were not selected and archiving of the file.

The selection phase is the one that varies most from one company to another: a bank may require several interviews and background checks, while a small shop may hire after a single interview. A good tool must therefore allow each company to **define its own stages**.

2.2 Business process modelling

A business process is a set of activities, performed by actors, that transforms inputs into a result of value for a customer. Business Process Management (BPM) studies how to identify, model, analyse, automate and improve such processes. The most common notations are **BPMN** (Business Process Model and Notation), the **UML activity diagram** and the **state machine diagram**.

A recruitment process lends itself well to modelling as a **state machine**: each application is an object whose state is the current stage, and the events (move, reject, hire) are transitions triggered by actors. Each transition can carry rules (who may perform it, which data are required) and effects (notification, history record). This is the approach retained for TalentFlow.

2.3 Personal data protection and CV security

A CV contains personal data: identity, contact details, photo, date of birth, career, sometimes salary expectations. In Cameroon, Law No. 2024/017 of 23 December 2024 on personal data protection sets rules for the processing of such data; in Europe, the General Data Protection Regulation (GDPR) plays the same role. The main principles common to these texts are:

- **lawfulness and consent:** the candidate must be informed of the purpose of the processing and agree to it;
- **purpose limitation and minimisation:** only the data needed for recruitment are collected;
- **limited retention:** data are kept for a defined period, then deleted or anonymised;
- **security:** the controller must protect data against loss, unauthorised access and disclosure;
- **rights of the data subject:** access, rectification and erasure.

On the technical side, the OWASP File Upload Cheat Sheet recommends validating the type of uploaded files with an allow-list and by their content (magic bytes), limiting their size, renaming them, storing them outside the web root, scanning them with an antivirus and serving them only to authorised users.

2.4 Web technologies

A modern web application is usually built in three tiers: a **client** running in the browser (single page application), an **application server** exposing a REST API, and a **database**. The proposed stack is the following:

- **React** is a JavaScript library for building user interfaces from reusable components; it is maintained by Meta and has a very large ecosystem.
- **Node.js** is a JavaScript runtime that executes code on the server with an event-driven, non-blocking model well suited to I/O-intensive applications such as APIs; **Express** is its most widely used minimalist web framework.
- **PostgreSQL** is an open-source relational database system known for its reliability, its compliance with SQL standards and advanced features such as full-text search, JSON types and arrays.

2.5 Comparison of possible solutions

Several solutions can be considered to solve the problem of application follow-up. Table 4 compares them.

Table 4: Comparison of the possible solutions

Solution	Advantages	Limits
Shared e-mail box + Excel	Free, known by everyone, no installation.	No single view, manual copying, no history, CVs scattered, no security, no notification.
Generic tools (Trello, Google Forms + Sheets)	Cheap, quick to set up, Kanban view available.	Not designed for recruitment, CVs stored on third-party servers, weak access control, no candidate communication.

Solution	Advantages	Limits
International SaaS ATS (e.g. Workable, Lever, Recruitee)	Complete features, mature, maintained.	Monthly subscription in foreign currency per user, international card payment required, data hosted abroad, interface not adapted to local channels.
Open-source ATS (e.g. OpenCATS)	Free licence, self-hosted.	Old technology and interface, little maintenance, customisation of stages limited, needs strong technical skills.
Custom platform (TalentFlow)	Adapted to local practices, data hosted under the company's control, customizable process, modern stack, low cost.	Development and maintenance effort; must be secured carefully.

The custom solution is retained because it is the only one that combines low cost, control of personal data and adaptation to the practices of Cameroonian SMEs, while remaining a realistic project for the duration of the internship.

Conclusion

This chapter presented the recruitment process, business process modelling, the requirements of personal data protection and the technologies used, and justified the choice of a custom platform. The next part is devoted to the empirical study.

PART TWO: EMPIRICAL STUDY

Chapter 3: Diagnosis of the existing system and requirements specification

Chapter 4: Choice and implementation of the solution

CHAPTER 3: DIAGNOSIS OF THE EXISTING SYSTEM AND REQUIREMENTS SPECIFICATION

Introduction

This chapter describes the recruitment process as it was observed on the field, explains how it works, identifies its strengths and its limits, and finally presents the requirements specification (cahier des charges) of the platform to be built.

3.1 Presentation of the existing system

The existing system is not a software application but a set of manual practices and general-purpose tools. Figure 2 shows its topology.

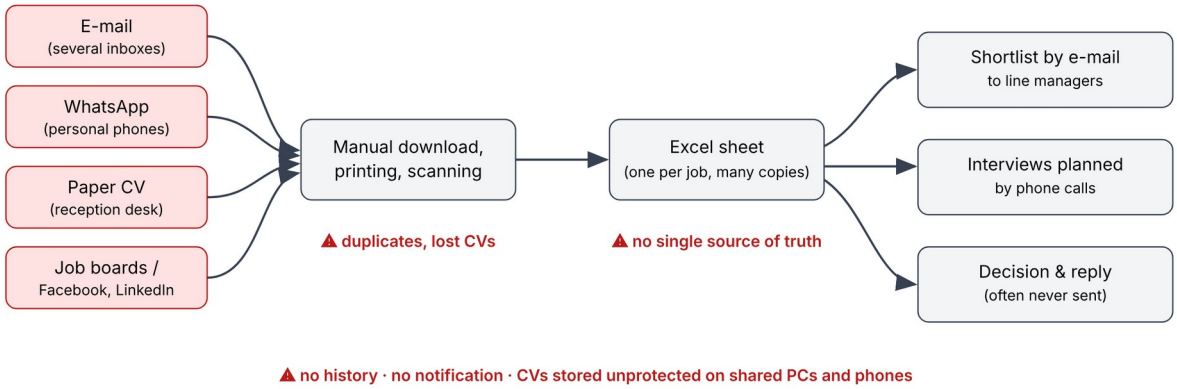


Figure 2: Topology of the existing recruitment process

3.2 Functioning of the existing system

1. A need is expressed by a manager; the HR officer writes the offer in Word and shares it by e-mail, on social networks and on WhatsApp groups.
2. Candidates send their CV to one of several e-mail addresses, to the personal WhatsApp number of an employee, or drop a paper copy at the reception.
3. The HR officer downloads, prints or scans the CVs, then records each candidate in an Excel sheet (name, phone, position, date).
4. Selected CVs are forwarded by e-mail to the line manager, who answers (or not) with comments.
5. Interviews are planned by phone; results are written on paper or in the Excel sheet.
6. The selected candidate is called; the other candidates generally receive no answer.

3.3 Advantages and limits of the existing system

Table 5: Strengths and weaknesses of the existing system

Aspect	Positive aspects	Limits
Technical	Tools already available and known by the staff; no installation.	Multiple sources, duplicates and lost CVs; no search in CV content; no history of decisions; file versions diverge.
Organisational	Flexible, adaptable to any situation.	Depends on one person; no shared view between HR and managers; candidates not informed; long delays.
Security	—	CVs stored on personal phones, mailboxes and shared computers; no access control; no traceability; no retention rule.
Financial	No direct cost.	Hidden cost: time spent on copying, searching and calling (estimated at several hours per week per recruitment); cost of bad or late hires.

The main limits that the new solution must overcome are therefore: **the dispersion of applications, the absence of a shared and traceable process, the lack of communication with candidates and the insecure storage of CVs**.

3.4 Requirements specification (cahier des charges)

3.4.1 Context and objectives of the project

Project: TalentFlow – web platform for recruitment and job application tracking. **Project owner (student):** SOLI MOKI Limnyuy Ngalla. **Academic supervisor:** TANDAH MARCELLE. **Field:** Employment / Human Resources. **Level of difficulty:** medium to advanced.

Problem: small companies receive CVs through several channels and lose track of applications. **Objective:** provide a single platform that centralises offers and applications, supports a customizable recruitment process, and handles CVs securely.

3.4.2 Scope

Included: job offer publication and public career page; online application with CV upload; manual import of applications received by e-mail or on paper; customizable stages; Kanban pipeline; candidate search and filters; evaluations and comments; e-mail and in-app notifications; HR dashboard; user and role management; secure CV storage. **Excluded from this version:** payroll, onboarding and employee management, video interviews, automatic CV parsing by artificial intelligence (planned as a perspective).

3.4.3 Actors

Table 6: Actors of the system and their roles

Actor	Description	Main rights
Visitor	Any Internet user consulting the career page.	Browse and search open offers.
Candidate	Person applying for an offer.	Apply with CV, track the status of the application, exercise the right to erasure.
Recruiter / Line manager	Employee involved in the selection for a given offer.	View the candidates of assigned offers, evaluate, comment, move candidates between stages.
HR Manager	Person responsible for recruitment in the company.	All recruiter rights + create/publish/close offers, configure stages and templates, dashboard, exports.
Administrator	Technical or company administrator.	Manage users, roles, company settings, retention policy, consult audit logs.

3.4.4 Functional requirements

The functional requirements are prioritised with the MoSCoW method (M = Must, S = Should, C = Could).

Table 7: Functional requirements of TalentFlow

Ref.	Requirement	Prio.
FR-01	Users authenticate with e-mail and password; accounts are linked to one company and one role.	M
FR-02	The HR manager creates, edits, publishes, closes and duplicates job offers (title, description, location, contract, deadline, skills, screening questions).	M
FR-03	Each published offer is visible on a public career page with a shareable link.	M
FR-04	A candidate applies online by filling a form and uploading a CV (PDF or DOCX, max 5 MB) and gives consent to data processing.	M
FR-05	HR staff can manually add an application received by e-mail, WhatsApp or on paper, indicating its source.	M
FR-06	Duplicate applications (same e-mail for the same offer) are detected and merged.	S
FR-07	The HR manager defines recruitment stages (add, rename, reorder, delete) per company template and per offer.	M
FR-08	Applications are displayed in a Kanban board; a candidate is moved between stages by drag and drop or by button.	M
FR-09	Every stage change is recorded with author and date (history).	M
FR-10	Users rate and comment candidates; comments are visible to the recruitment team only.	S
FR-11	Candidates are searched by keywords, including the text content of the CV.	M
FR-12	Candidates are filtered by offer, stage, experience, skills, source, rating and date.	M
FR-13	An e-mail is sent to the candidate at application and at configurable stage changes (templates with variables).	M

Ref.	Requirement	Prio.
FR-14	Users receive in-app notifications (new applications, pending reviews, upcoming deadlines).	S
FR-15	The HR dashboard shows open offers, applications per period, funnel by stage, sources and time-to-hire.	M
FR-16	Candidates can follow the status of their application through a private link.	S
FR-17	Data of a candidate can be exported or anonymised on request (right to erasure).	S
FR-18	Applications older than the retention period are anonymised automatically.	C
FR-19	Lists of candidates can be exported in CSV/Excel.	C
FR-20	The administrator manages users and roles and consults the audit log.	M

3.4.5 Non-functional requirements

Table 8: Non-functional requirements

Category	Requirement
Security	HTTPS only; passwords hashed with bcrypt; JWT access tokens (15 min) + refresh token in httpOnly cookie; RBAC on every endpoint; CVs encrypted at rest (AES-256-GCM) and served through signed URLs valid 5 minutes; antivirus scan of every upload; audit log of every CV access; protection against OWASP Top 10 (injection, XSS, CSRF, broken access control).
Privacy	Explicit consent at application; retention period configurable (24 months by default); right of access and erasure.
Performance	API response time < 500 ms for 95 % of requests; search on 10,000 candidates < 1 s; upload of a 5 MB CV < 5 s on a 4G connection.
Availability	Target availability of 99 %; daily database backup kept 30 days.
Usability	Responsive interface usable on smartphone (career page) and desktop (back-office); bilingual French/English; a new HR user can publish an offer in less than 10 minutes without training.
Compatibility	Latest versions of Chrome, Firefox, Edge and Safari; Android and iOS browsers.
Maintainability	Code versioned with Git; layered architecture; automated tests on critical functions; documented API.
Portability	Deployment with Docker on any Linux server or cloud provider.

3.4.6 Constraints

- **Technical:** React for the front-end, Node.js/Express for the back-end, PostgreSQL for the database (stack imposed by the project).
- **Time:** the project must be designed, developed and documented within the nine weeks of the internship.
- **Budget:** use of free and open-source tools; low hosting cost suited to an SME.
- **Legal:** compliance with the rules on personal data protection.

3.4.7 Deliverables and acceptance criteria

Table 9: Deliverables of the project

Deliverable	Acceptance criteria
Requirements specification and UML design	Validated by the professional and academic supervisors.
Source code (front-end, back-end, database scripts)	Versioned in Git; builds and runs with a single Docker Compose command.
Deployed test instance	All “Must” requirements demonstrated on a realistic scenario.
Test report	Critical tests (authentication, upload, stage transition, access control) pass.
User guide and technical documentation	An HR user can publish an offer and process applications using the guide.
Internship report and presentation	Conform to the SEAS-IUC report format.

3.4.8 Planning

The project was planned over the nine weeks of the internship as shown in Figure 3.

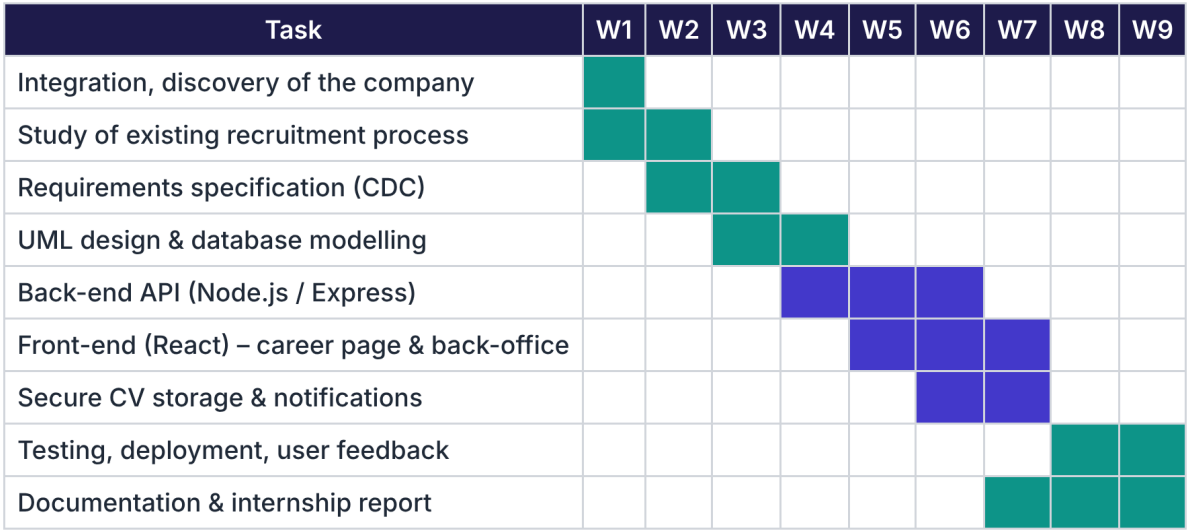


Figure 3: Gantt chart of the project

Conclusion

The diagnosis showed that the existing process, based on general-purpose tools, leads to lost applications, lack of traceability and insecure storage of personal data. The requirements specification translated these findings into precise functional and non-functional requirements. The next chapter presents how these requirements were implemented.

CHAPTER 4: CHOICE AND IMPLEMENTATION OF THE SOLUTION

Introduction

This chapter presents the technical notebook of the new system: the choice and justification of its components, its architecture, its UML design, the modelling of the recruitment process, the security of CVs, the installation and configuration procedure, the interfaces obtained, the tests carried out and the cost estimate.

4.1 Technical notebook of the new system

4.1.1 Choice and justification of the components

Table 10: Choice and justification of the technologies

Component	Choice	Justification
Front-end	React 18 + Vite, React Router, TanStack Query, Tailwind CSS	Component-based, very large community, fast development; Vite gives instant builds; TanStack Query handles caching and synchronisation with the API.
Drag and drop	dnd-kit	Accessible, lightweight library for the Kanban board.
Back-end	Node.js 20 LTS + Express 4	Same language (JavaScript) on client and server; non-blocking I/O suited to file uploads and notifications; simple and well-known framework.
Validation	Zod	Single schema used to validate requests and to type data.
ORM	Prisma	Typed queries, migrations versioned with the code, protection against SQL injection.
Database	PostgreSQL 16	Reliable relational DBMS, transactions, enumerated types, arrays, native full-text search (tsvector) used to search CV content.
File storage	MinIO (S3-compatible), private bucket	Files kept outside the web server, pre-signed URLs, can be replaced by any S3 provider.
Antivirus	ClamAV (clamd)	Free, open-source antivirus engine usable as a network service.
Queue & cache	Redis + BullMQ	Asynchronous sending of e-mails and scheduled jobs (reminders, retention).
E-mail	Nodemailer + SMTP	Standard, works with any e-mail provider.
Real-time	Socket.IO	Instant in-app notifications and Kanban updates.
Deployment	Docker Compose + Nginx + Let’s Encrypt	Reproducible installation, free TLS certificates.

4.1.2 Architecture of the solution

The platform follows a three-tier architecture (Figure 4). The React single page application communicates with the Express API over HTTPS using JSON. The API applies authentication,

authorisation and validation, then accesses PostgreSQL through Prisma and the object storage for CV files. A worker process consumes the Redis queue to send e-mails and run scheduled jobs.

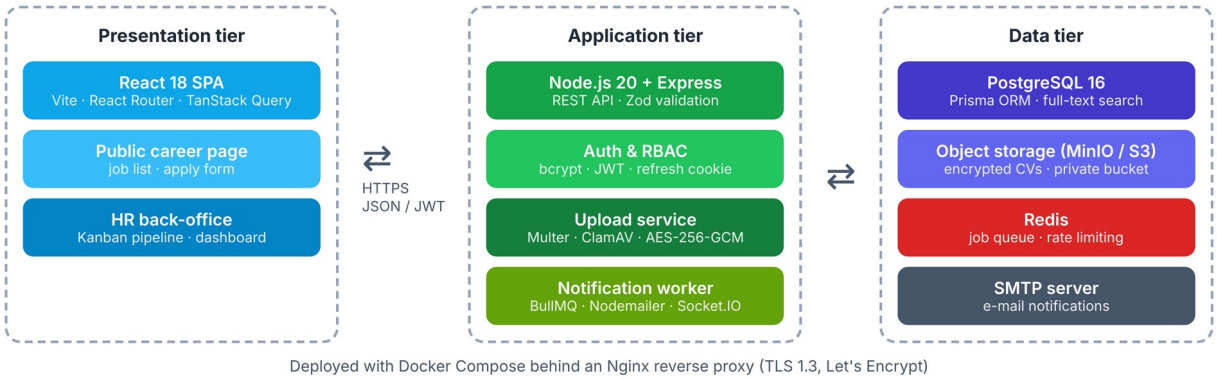


Figure 4: Three-tier architecture of TalentFlow

The back-end code is organised in layers: **routes** (HTTP endpoints) → **middlewares** (authentication, RBAC, validation, upload) → **services** (business rules) → **repositories** (Prisma). This separation makes the business rules independent of the framework and easier to test.

4.2 UML design of the system

4.2.1 Use case diagram

The use case diagram (Figure 5) shows the five actors and the main services they expect from the system.

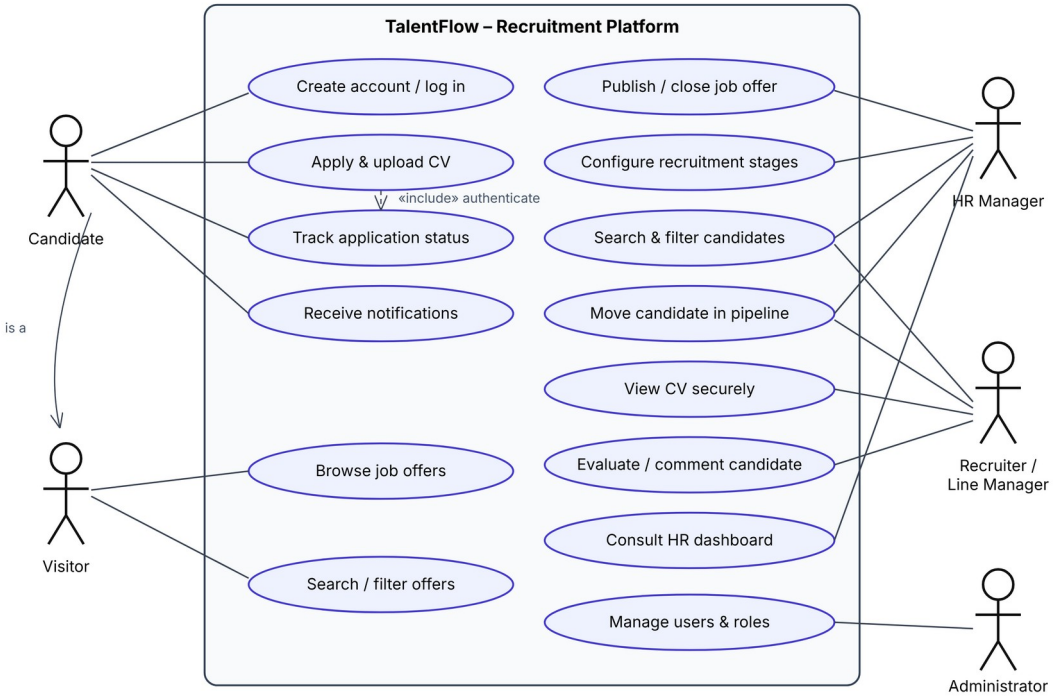


Figure 5: Use case diagram

Table 11: Textual description of the use case "Apply & upload CV"

Item	Description
Actor	Candidate
Pre-condition	The job offer is published and its deadline is not passed.
Nominal scenario	1. The candidate opens the offer page and clicks "Apply". 2. The system displays the form and the screening questions. 3. The candidate fills the form, attaches the CV and accepts the privacy notice. 4. The system validates the file, scans it, encrypts and stores it. 5. The system creates the application in the first stage and sends a confirmation e-mail.
Alternative scenarios	4a. Invalid type, size > 5 MB or virus detected: the file is rejected with an explicit message. 5a. An application with the same e-mail already exists for this offer: the CV is updated instead of creating a duplicate.
Post-condition	The application appears in the "New" column of the pipeline and HR users are notified.

4.2.2 Class diagram

The class diagram (Figure 6) shows the main entities. A **Company** publishes **JobOffers**; each offer owns an ordered list of **PipelineStages**. An **Application** links a **Candidate** to an offer and points to its current stage; each change of stage creates a **StageHistory** entry. The CV is represented by a **CvDocument** that stores only the key of the encrypted file and its fingerprint.

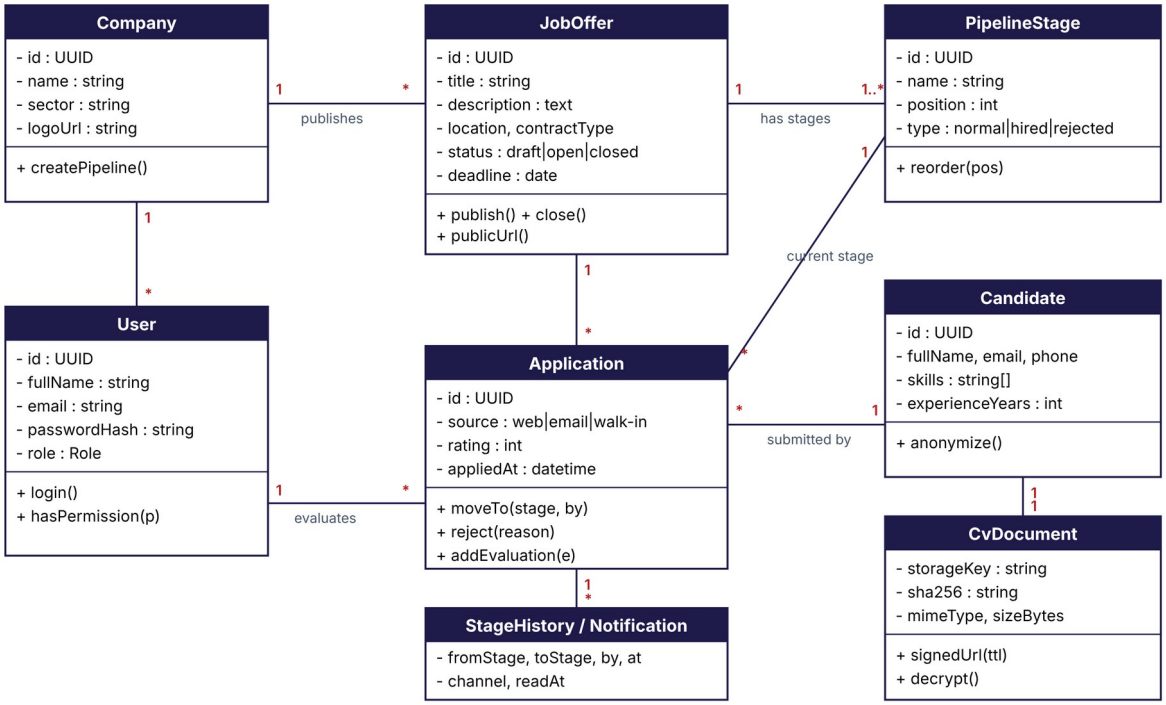


Figure 6: Class diagram

4.2.3 Activity diagram

The activity diagram (Figure 7) describes the life cycle of an application, from the publication of the offer to the final decision, across the three swimlanes candidate, system and HR.

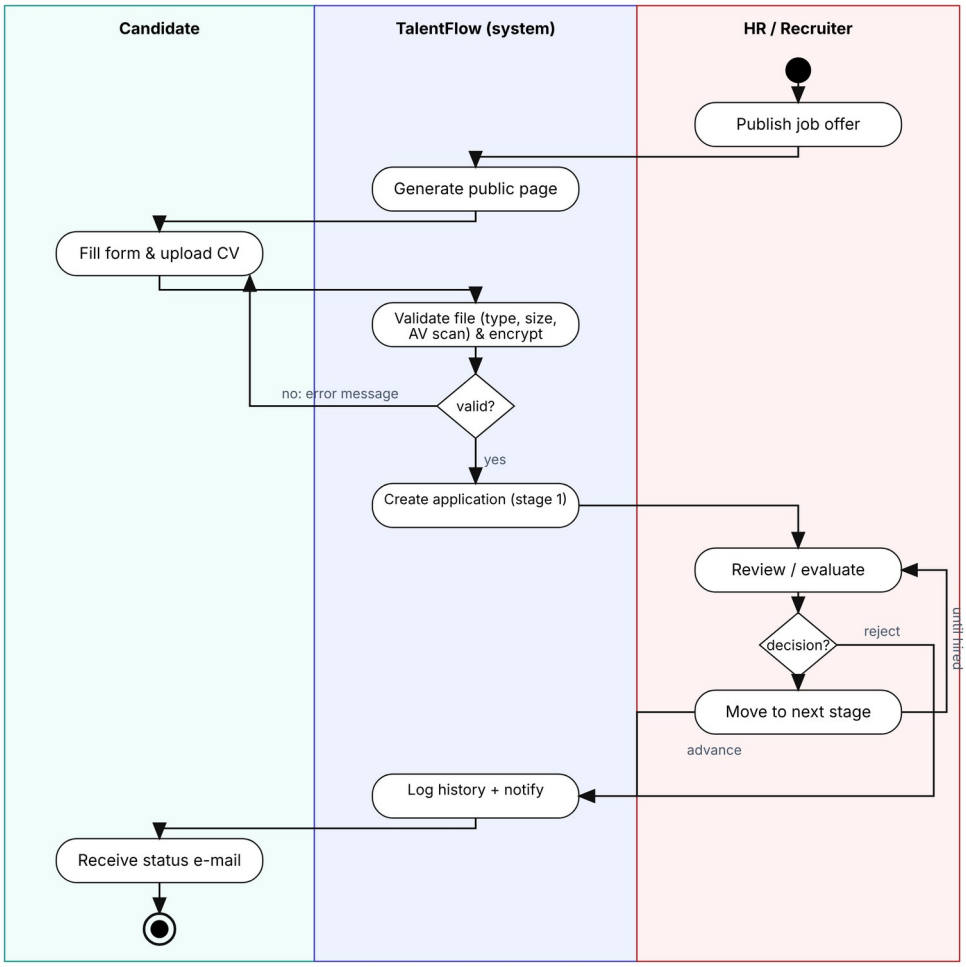


Figure 7: Activity diagram of the processing of an application

4.2.4 Sequence diagram

The sequence diagram (Figure 8) details the secure upload of a CV and its later consultation by an HR user.

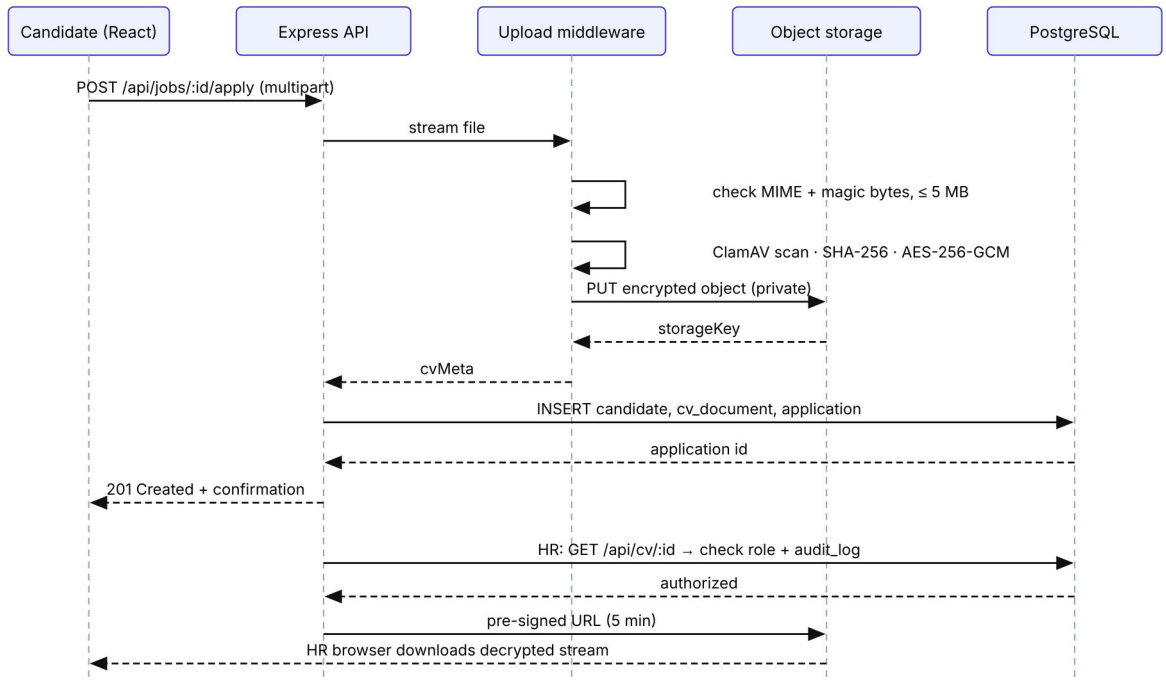


Figure 8: Sequence diagram of the secure upload and consultation of a CV

4.3 Modelling of the recruitment business process

The heart of the platform is the modelling of the recruitment process as a **configurable state machine** (Figure 9). Instead of hard-coding the stages, they are stored as data in the table `pipeline_stages`, with a position and a kind (`normal`, `hired` or `rejected`). When an offer is created, the stages of the company template are copied, and the HR manager can then adapt them to the offer.

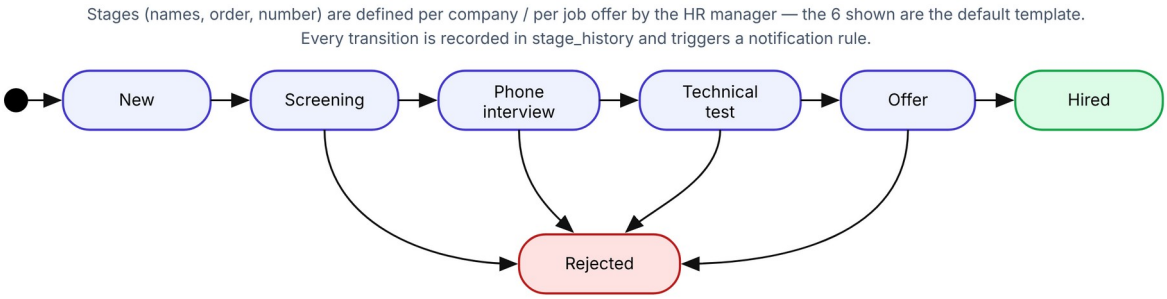


Figure 9: State machine of an application with the default stages

The following business rules are enforced by the service layer:

- **BR1** – An application is always in exactly one stage of the offer it belongs to.
- **BR2** – Only users with the role HR manager or recruiter assigned to the offer can move an application.
- **BR3** – Moving to a stage of kind `rejected` requires a reason; the candidate is notified after a configurable delay (48 h by default) to avoid abrupt messages.

- **BR4** – Moving to `hired` closes the application; when the number of positions is reached, the offer can be closed automatically.
- **BR5** – A stage that still contains applications cannot be deleted; its applications must first be moved.
- **BR6** – Every transition is written in `stage_history` in the same database transaction as the update of the application, and triggers the notification rules attached to the target stage.

The code below shows how a transition is executed inside a transaction, so that the application and its history can never be inconsistent:

```
// services/application.service.js
async function moveToStage({ applicationId, toStageId, user, reason }) {
  return prisma.$transaction(async (tx) => {
    const app = await tx.application.findUniqueOrThrow({
      where: { id: applicationId }, include: { stage: true } });
    const target = await tx.pipelineStage.findFirstOrThrow({
      where: { id: toStageId, jobOfferId: app.jobOfferId } });   // BR1
    await assertCanManageOffer(tx, user, app.jobOfferId);          // BR2
    if (target.kind === 'rejected' && !reason)
      throw new BusinessError('A rejection reason is required'); // BR3
    const updated = await tx.application.update({
      where: { id: applicationId }, data: { stageId: target.id } });
    await tx.stageHistory.create({ data: {                         // BR6
      applicationId, fromStageId: app.stageId, toStageId: target.id,
      movedById: user.id, reason } });
    await notificationQueue.add('stage-changed',
      { applicationId, toStageId: target.id }, { delay: target.kind === 'rejected' ? 48 *
3600e3 : 0 });
    return updated;
  });
}
```

4.4 Database design

The physical data model (Figure 10) is implemented in PostgreSQL. Primary keys are UUIDs, which avoids exposing sequential identifiers in URLs. Foreign keys guarantee referential integrity; a unique constraint on (job_offer_id, candidate_id) prevents duplicate applications; a `tsvector` column indexed with GIN allows full-text search in offers and in the text extracted from CVs.

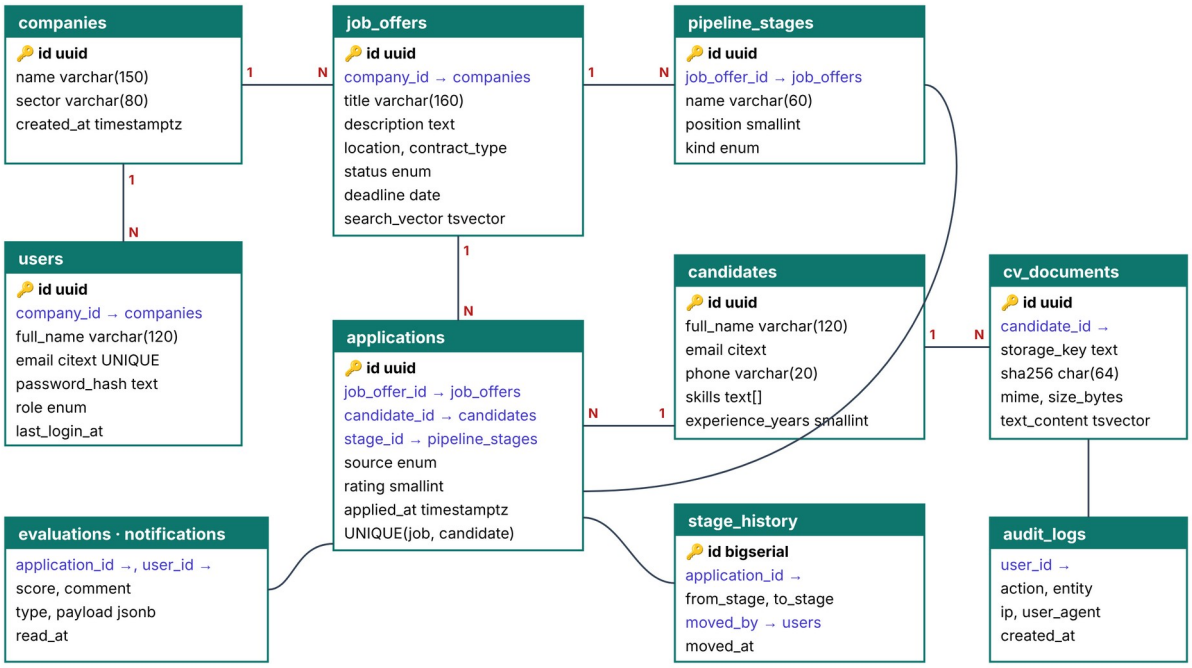


Figure 10: Physical data model (PostgreSQL)

4.5 Secure management of CVs

Because CVs are the most sensitive data of the platform, they are protected by several successive layers of defence (Figure 11).

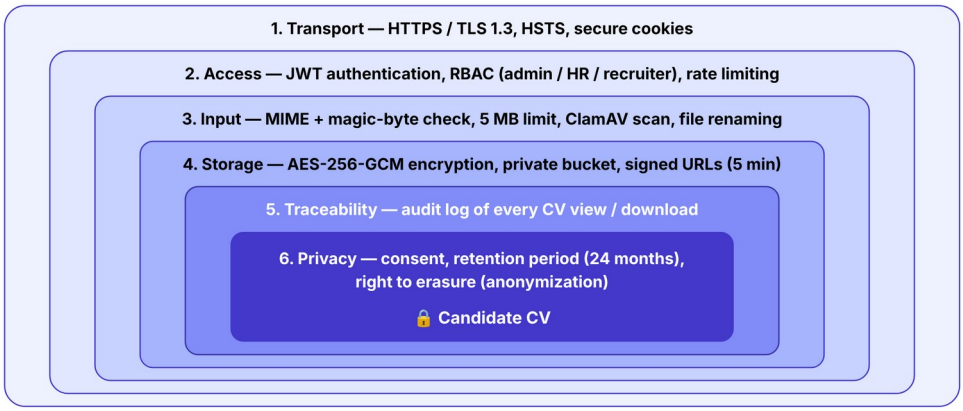


Figure 11: Defence-in-depth layers protecting the CVs

Table 12: Security measures applied to CVs

Threat	Measure implemented
Upload of a malicious file (script, macro, virus)	Allow-list of extensions (.pdf, .docx) and verification of the real type with magic bytes; size limit of 5 MB; ClamAV scan; random file name; files never served from the web server.
Theft of files from the server or the storage	Encryption with AES-256-GCM before storage, with a key kept in the server environment and never in the database; private bucket.

Threat	Measure implemented
Unauthorised access to a CV	Authentication by JWT; role-based access control and check that the user belongs to the company and is assigned to the offer; pre-signed URL valid for 5 minutes only.
Silent abuse by an internal user	Audit log of every CV view and download (user, date, IP address); watermark on downloaded copies.
Interception on the network	HTTPS only (TLS 1.2+), HSTS, secure and httpOnly cookies.
Brute force on accounts	Password hashing with bcrypt (cost 12); rate limiting; account locked for 15 minutes after 5 failures.
Excessive retention	Consent at application; retention period of 24 months by default, after which a scheduled job anonymises the candidate and deletes the file; erasure on request.

The upload middleware below illustrates the first layers of this protection:

```
// middlewares/cvUpload.js
const upload = multer({ storage: multer.memoryStorage(),
  limits: { fileSize: 5 * 1024 * 1024, files: 1 } });
const ALLOWED = { 'application/pdf': 'pdf',
  'application/vnd.openxmlformats-officedocument.wordprocessingml.document': 'docx' };

async function secureCv(req, res, next) {
  const file = req.file;
  if (!file) return res.status(400).json({ error: 'CV is required' });
  const type = await fileTypeFromBuffer(file.buffer);           // magic bytes
  if (!type || !ALLOWED[type.mime]) return res.status(415).json({ error: 'PDF or DOCX
only' });
  const scan = await clamav.scanBuffer(file.buffer);
  if (scan.isInfected) return res.status(422).json({ error: 'File rejected' });
  const iv = crypto.randomBytes(12);
  const cipher = crypto.createCipheriv('aes-256-gcm', CV_KEY, iv);
  const enc = Buffer.concat([cipher.update(file.buffer), cipher.final()]);
  const key = `cv/${crypto.randomUUID()}.${ALLOWED[type.mime]}.enc`;
  await storage.putObject(BUCKET, key, Buffer.concat([iv, cipher.getAuthTag(), enc]));
  req.cv = { storageKey: key, mime: type.mime, size: file.size,
    sha256: crypto.createHash('sha256').update(file.buffer).digest('hex') };
  next();
}
```

4.6 REST API

Table 13: Main endpoints of the REST API

Method & route	Description	Access
POST /api/auth/login	Authenticate and obtain tokens	Public
POST /api/auth/refresh	Renew the access token	Cookie
GET /api/public/jobs	List open offers (search, filters)	Public
POST /api/public/jobs/:id/apply	Apply with CV (multipart)	Public
GET /api/public/applications/:token	Candidate follows the status	Private link

Method & route	Description	Access
GET | POST /api/jobs	List / create offers	HR
PATCH /api/jobs/:id	Edit, publish, close an offer	HR
GET | PUT /api/jobs/:id/stages	Read / reorder the stages	HR
GET /api/jobs/:id/pipeline	Kanban data	HR, Recruiter
POST /api/applications/:id/move	Move to a stage	HR, Recruiter
POST /api/applications/:id/evaluations	Rate / comment	HR, Recruiter
GET /api/candidates	Search and filter candidates	HR, Recruiter
GET /api/cv/:id/url	Signed URL to view a CV (logged)	HR, Recruiter
DELETE /api/candidates/:id	Anonymise (right to erasure)	HR, Admin
GET /api/dashboard	Indicators of the HR dashboard	HR
GET | POST /api/users	Manage users and roles	Admin
GET /api/audit-logs	Consult the audit trail	Admin

4.7 Installation and configuration procedure

4.7.1 Prerequisites

- ➢ A Linux server (Ubuntu 22.04 LTS) with at least 2 vCPU, 4 GB RAM and 40 GB of disk;
- ➢ Docker Engine 24+ and Docker Compose v2;
- ➢ A domain name pointing to the server (e.g. recrutement.company.cm);
- ➢ An SMTP account for sending e-mails.

4.7.2 Installation steps

```
# 1. Get the source code
git clone https://github.com/<organisation>/talentflow.git && cd talentflow
# 2. Create the configuration file
cp .env.example .env && nano .env
# 3. Generate the CV encryption key (32 bytes, base64)
openssl rand -base64 32
# 4. Build and start all services
docker compose up -d --build
# 5. Create the database schema and the first administrator
docker compose exec api npx prisma migrate deploy
docker compose exec api npm run seed:admin -- --email admin@company.cm
# 6. Obtain the TLS certificate
docker compose run --rm certbot certonly --webroot -w /var/www/certbot -d
recrutement.company.cm
```

4.7.3 Configuration

```
# .env (extract)
NODE_ENV=production
APP_URL=https://recrutement.company.cm
DATABASE_URL=postgresql://talentflow:********@db:5432/talentflow
JWT_ACCESS_SECRET=********          # 64 random characters
JWT_ACCESS_TTL=15m
CV_ENCRYPTION_KEY=********          # output of: openssl rand -base64 32
S3_ENDPOINT=http://minio:9000
S3_BUCKET=talentflow-cv
CLAMAV_HOST=clamav
REDIS_URL=redis://redis:6379
SMTP_HOST=smtp.provider.com
SMTP_USER=no-reply@company.cm
CV_RETENTION_MONTHS=24
SIGNED_URL_TTL_SECONDS=300
```

After installation, the administrator logs in, fills in the company profile (name, logo, colours of the career page), creates the user accounts of the HR team, defines the default recruitment stages and adapts the e-mail templates. A daily backup of the database is configured with a cron job running `pg_dump`.

4.8 Presentation of the interfaces

This section presents the main screens of the platform.

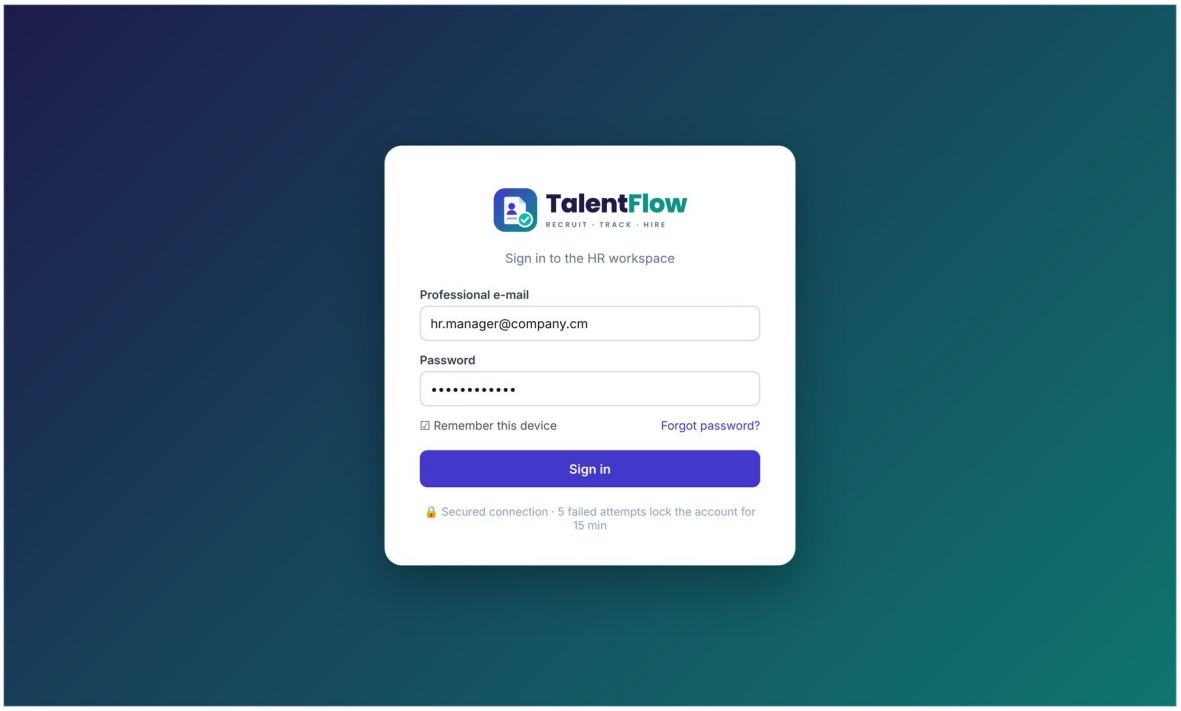


Figure 12: Login page of the HR workspace

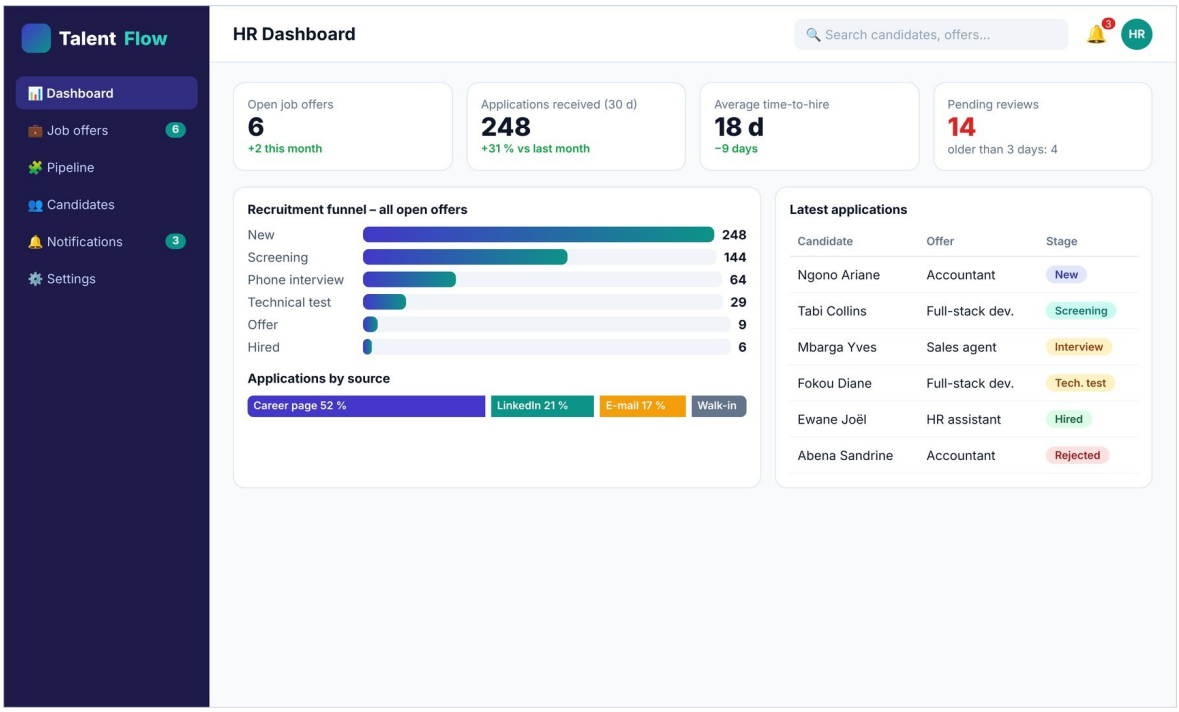


Figure 13: HR dashboard

The HR dashboard (Figure 13) gives at a glance the number of open offers, the applications received, the average time-to-hire and the reviews waiting for too long, as well as the recruitment funnel and the distribution of applications by source.

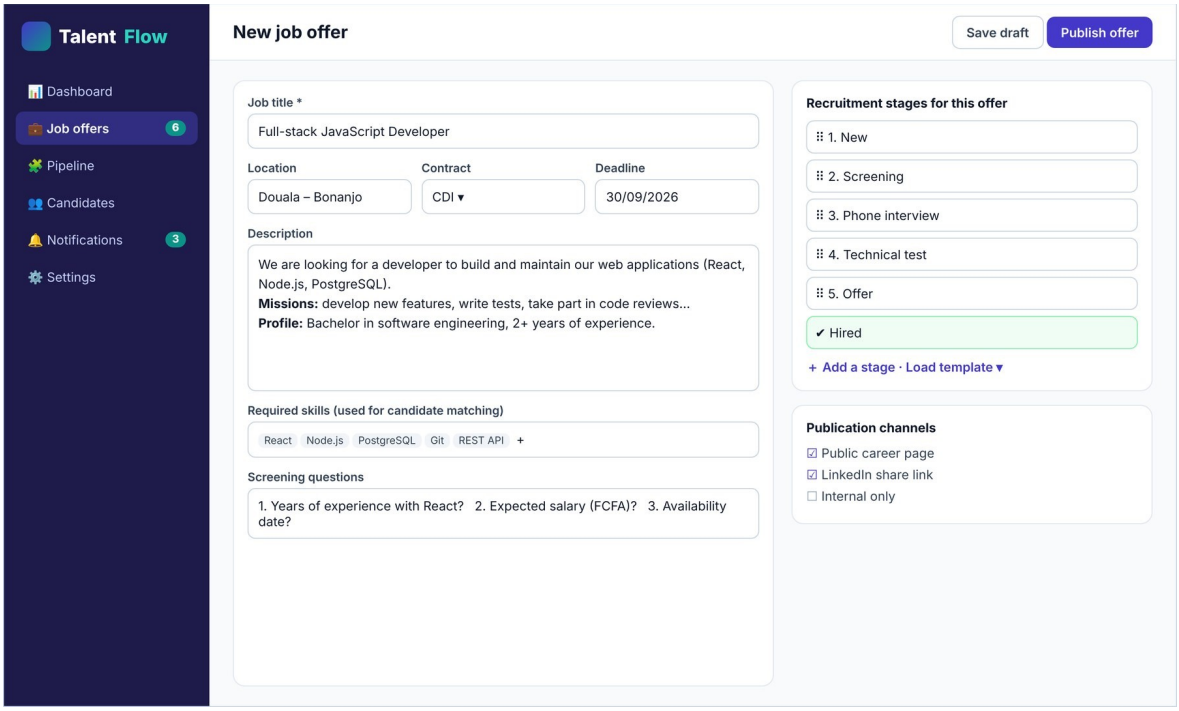


Figure 14: Creation of a job offer with its customizable stages

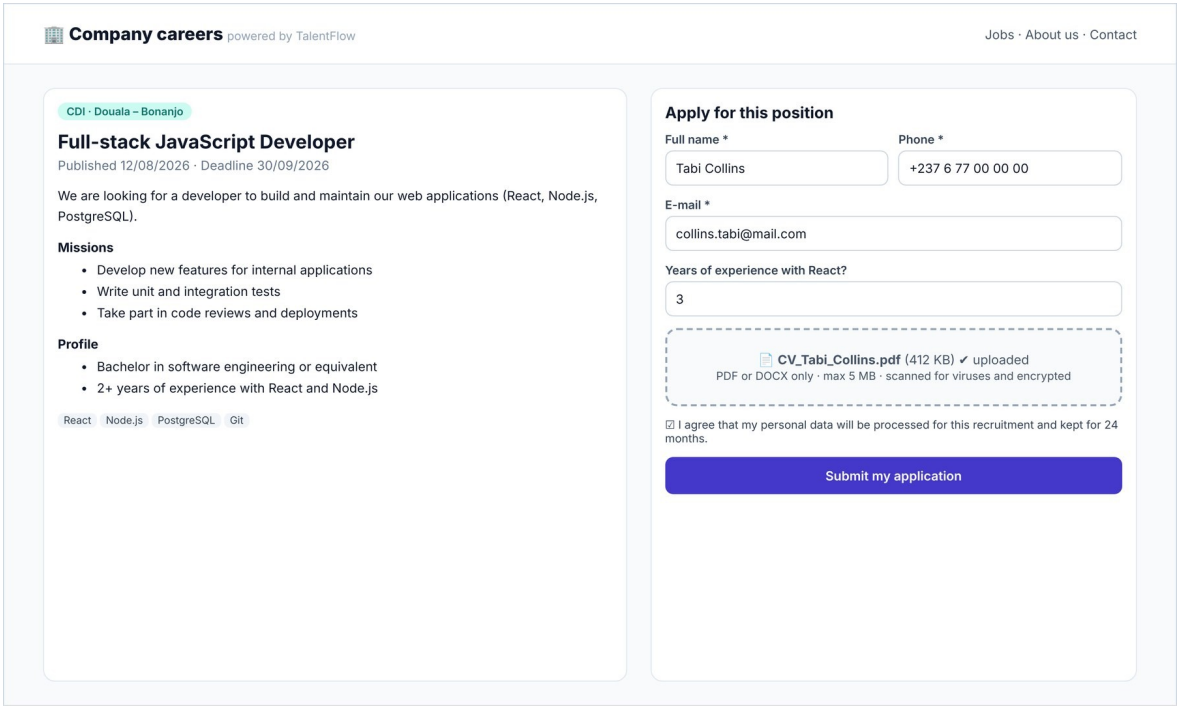


Figure 15: Public career page and application form with secure CV upload

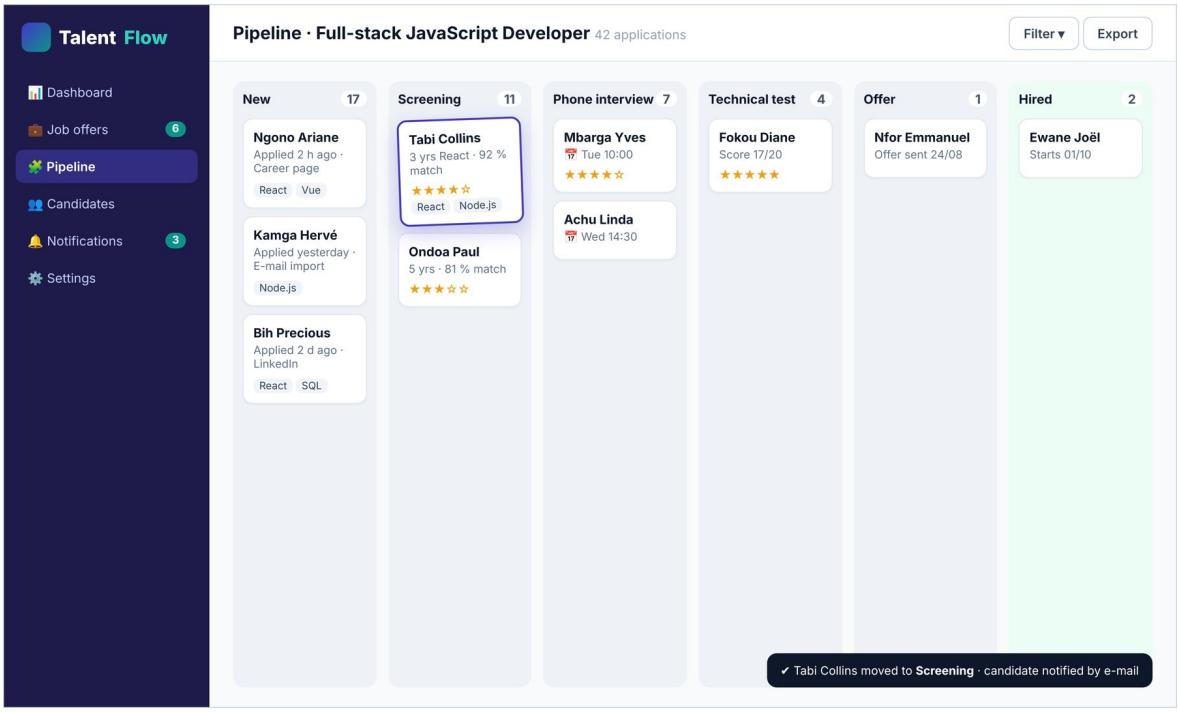


Figure 16: Kanban board of the recruitment pipeline

On the Kanban board (Figure 16), the recruiter moves a candidate from one stage to another by drag and drop. The move is recorded in the history and the candidate automatically receives the e-mail configured for the new stage.

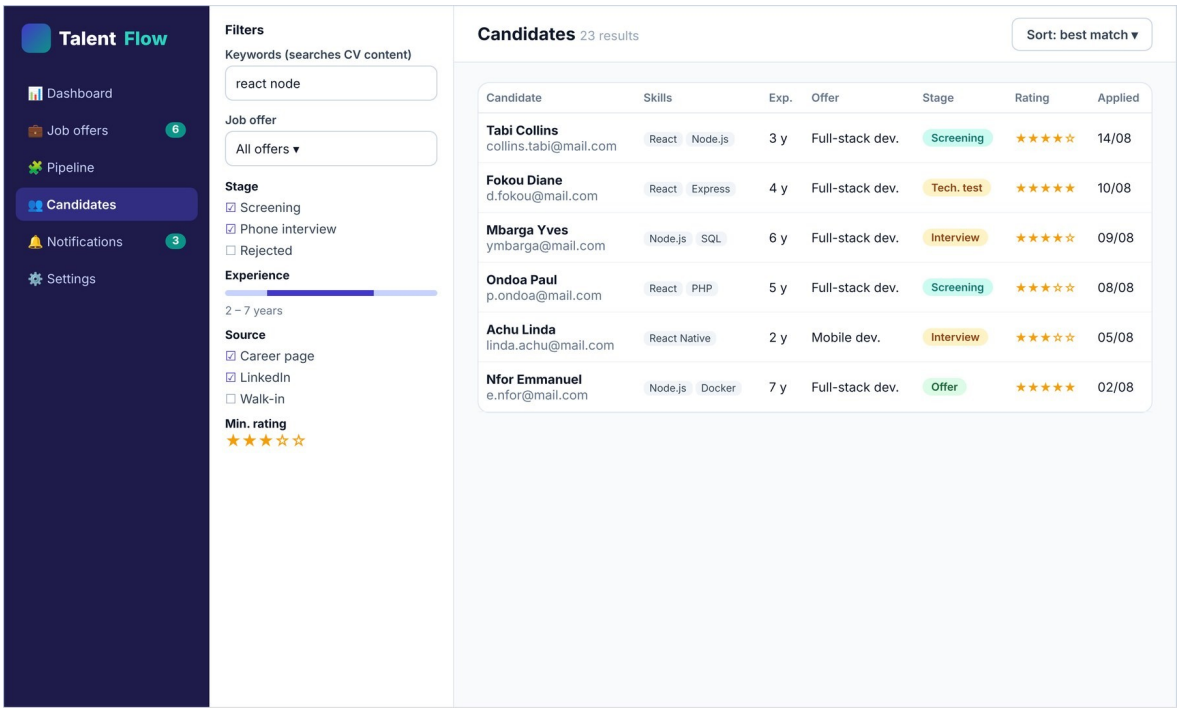


Figure 17: Candidate search with filters

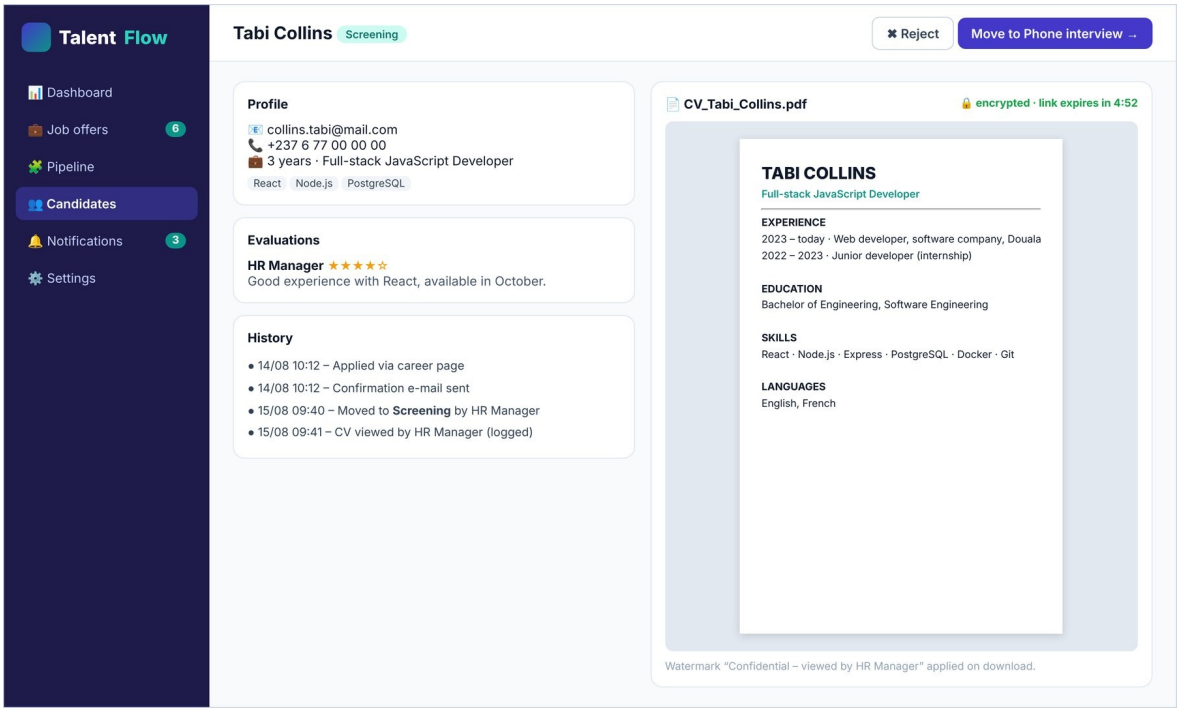


Figure 18: Candidate profile with secure CV viewer and history

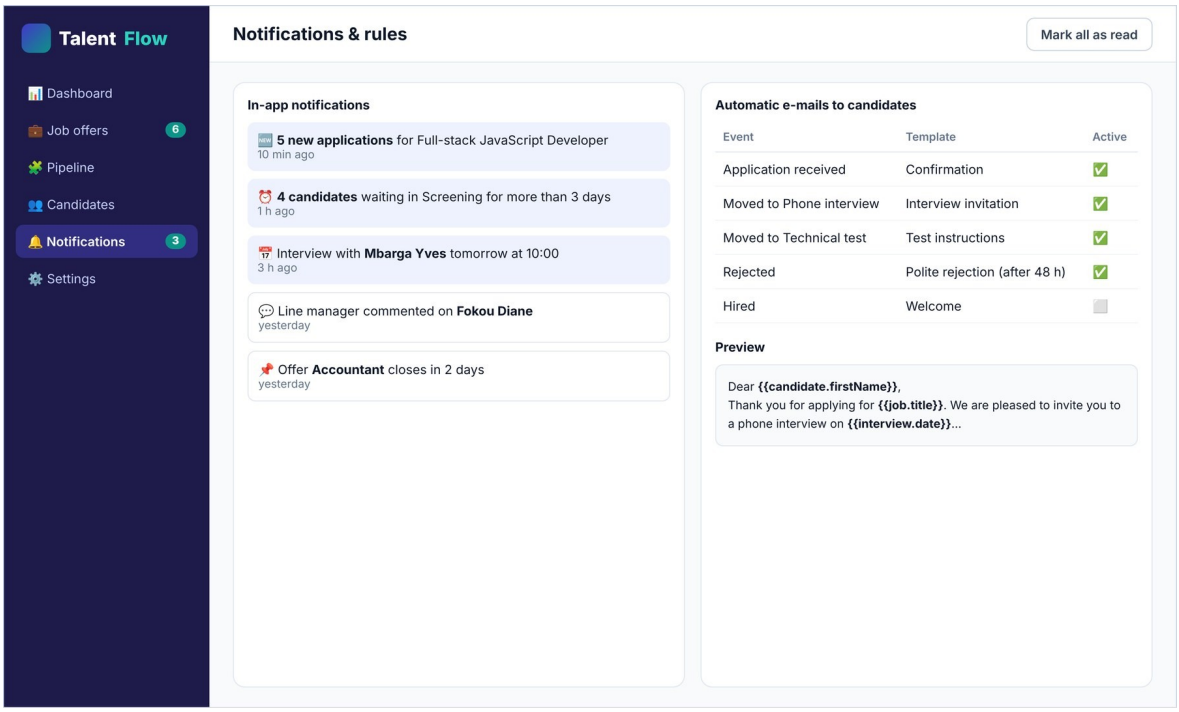


Figure 19: Notifications and automatic e-mail rules

4.9 Tests

Unit tests were written with **Jest** for the business services and integration tests with **Supertest** for the API endpoints, using a dedicated PostgreSQL test database. Table 14 summarises the main test cases.

Table 14: Main test cases

ID	Test case	Expected result	Status
T01	Login with a wrong password 5 times	Account locked for 15 minutes	Passed
T02	Apply with a PDF of 2 MB	Application created in stage “New”, confirmation e-mail queued	Passed
T03	Apply with an .exe renamed as .pdf	HTTP 415 – file rejected	Passed
T04	Apply with the EICAR antivirus test file	HTTP 422 – file rejected	Passed
T05	Apply twice with the same e-mail for the same offer	No duplicate; CV updated	Passed
T06	Recruiter moves a candidate of an offer not assigned to them	HTTP 403 – forbidden	Passed
T07	Reject a candidate without reason	HTTP 400 – reason required	Passed
T08	Open the signed URL of a CV after 6 minutes	Access denied (link expired)	Passed
T09	Search “react node” among 10,000 generated candidates	Relevant results in less than 1 s	Passed
T10	Delete a stage that still contains applications	Operation refused with a message	Passed

4.10 Cost estimate of the solution

The cost estimate includes the material and services needed to run the platform for one year, and the handwork needed to develop and deploy it. Prices are indicative and given in CFA francs (FCFA).

Table 15: Cost estimate of the solution (FCFA)

Item	Quantity	Unit price	Total
Material and services			
Cloud server (VPS 2 vCPU, 4 GB RAM) – 12 months	12 months	15 000	180 000
Domain name (.cm) – 1 year	1	25 000	25 000
Transactional e-mail service – 12 months	12 months	5 000	60 000
TLS certificate (Let's Encrypt), open-source software	—	0	0
Subtotal material and services			**265 000**
Handwork			
Analysis and design (requirements, UML, database)	10 days	25 000	250 000
Development (back-end, front-end, security)	25 days	25 000	625 000
Tests, deployment and training of users	5 days	25 000	125 000
Subtotal handwork			**1 000 000**
TOTAL (material + handwork)			**1 265 000**

For comparison, an international SaaS applicant tracking system typically costs several hundred thousand FCFA per year for a small team, with data hosted abroad. Once developed, TalentFlow can be offered to several SMEs on the same infrastructure (multi-company model), which makes it even more economical.

Conclusion

This chapter presented the technical choices, the architecture, the UML design, the modelling of the recruitment process as a configurable state machine, the measures that protect the CVs, the installation procedure, the interfaces, the tests and the cost of the solution. TalentFlow meets the "Must" requirements of the specification.

GENERAL CONCLUSION

This internship at Afriland First Bank, Douala – Bonanjo, from July 1 to August 31, 2026, allowed me to discover the functioning of a large banking institution and to work on a concrete problem: **small companies receive CVs through several channels and lose track of applications**.

Summary of the problems solved. The study of the existing process showed that applications were dispersed between e-mails, WhatsApp, paper and Excel sheets, that no one had a shared view of the process, that candidates were not informed and that CVs were stored without protection. TalentFlow solves these problems by centralising offers and applications in a single platform, by giving a clear and traceable view of each application, by informing candidates automatically and by protecting CVs.

Methodology and proposed solution. After the analysis of the existing system and the writing of the requirements specification, the solution was modelled with UML, the recruitment process was represented as a configurable state machine, and the platform was developed with React, Node.js/Express and PostgreSQL. The secure management of CVs relies on a defence-in-depth approach: validation and antivirus scanning of files, encryption at rest, role-based access, short-lived signed links and audit logging.

Difficulties encountered and solutions. The main difficulties were the limited time to cover all the requirements (solved by prioritising them with the MoSCoW method), the variety of recruitment practices to be supported (solved by making the stages configurable instead of fixed), the complexity of file security (solved by following the OWASP recommendations step by step), and the integration of new technologies such as Prisma and BullMQ (solved through documentation and the advice of my supervisors).

Perspectives. Several improvements are planned for the next versions: automatic extraction of information from CVs and matching score with the offer using natural language processing; integration with WhatsApp Business and e-mail inboxes to import applications automatically; scheduling of interviews synchronised with calendars; a mobile application for recruiters; multi-company offer as a service for SME customers; and statistics on diversity and recruitment costs.

On a personal level, this internship strengthened my skills in requirements analysis, business process modelling, full-stack web development and application security, as well as my communication and teamwork skills in a professional environment.

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9. ClamAV documentation – https://docs.clamav.net
10. Afriland First Bank – https://www.afrilandfirstbank.com

APPENDIX

Appendix 1: Extract of the database creation script

```
CREATE TYPE stage_kind AS ENUM ('normal', 'hired', 'rejected');
CREATE TYPE user_role  AS ENUM ('admin', 'hr_manager', 'recruiter');

CREATE TABLE job_offers (
  id            uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  company_id    uuid NOT NULL REFERENCES companies(id),
  title         varchar(160) NOT NULL,
  description   text NOT NULL,
  status        varchar(10) NOT NULL DEFAULT 'draft',
  deadline      date,
  search_vector tsvector GENERATED ALWAYS AS
    (to_tsvector('french', title || ' ' || description)) STORED
);
CREATE INDEX job_offers_search_idx ON job_offers USING gin(search_vector);

CREATE TABLE pipeline_stages (
  id           uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  job_offer_id uuid NOT NULL REFERENCES job_offers(id) ON DELETE CASCADE,
  name         varchar(60) NOT NULL,
  position     smallint NOT NULL,
  kind         stage_kind NOT NULL DEFAULT 'normal',
  UNIQUE (job_offer_id, position) DEFERRABLE INITIALLY DEFERRED
);

CREATE TABLE applications (
  id           uuid PRIMARY KEY DEFAULT gen_random_uuid(),
  job_offer_id uuid NOT NULL REFERENCES job_offers(id),
  candidate_id uuid NOT NULL REFERENCES candidates(id),
  stage_id     uuid NOT NULL REFERENCES pipeline_stages(id),
  source       varchar(20) NOT NULL DEFAULT 'web',
  rating       smallint CHECK (rating BETWEEN 1 AND 5),
  applied_at   timestamptz NOT NULL DEFAULT now(),
  UNIQUE (job_offer_id, candidate_id)
);
```

Appendix 2: Quick user guide for the HR manager

1. Log in at the address of the platform with the account created by the administrator.
2. Open **Job offers** → **New offer**, fill in the form, adapt the stages if necessary and click **Publish**.
3. Share the public link of the offer on social networks and job boards.
4. Add manually the CVs received by e-mail or on paper with **Add application** and select the source.
5. Open **Pipeline**, review the candidates and move them between stages; rate and comment them.
6. Use **Candidates** to search by keyword or filter by stage, experience or rating.
7. Follow the indicators on the **Dashboard** and close the offer when the position is filled.